

Solutions

Question 1 – Solution

Step 1: Variables

- Doctors = {D1, D2, D3}
 - Shifts = {Morning, Afternoon, Night}
-

Step 2: Constraints

- D1 cannot work the Night shift.
 - D2 must work before D3 (Morning < Afternoon < Night).
 - Only one doctor per shift.
 - D3 cannot work the Morning shift.
 - Every doctor must be assigned exactly one shift.
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Step 3: Backtracking Search

Initial State

D1 = ?

D2 = ?

D3 = ?

Assign D1 = Morning

✓ Constraint satisfied

D1 = Morning

Remaining shifts:

- Afternoon
 - Night
-

Assign D2 = Afternoon

✓ Constraint satisfied

D2 = Afternoon

Remaining shift:

- Night

Assign D3 = Night

Constraint Check:

- D3 $\neq$ Morning ✓
- D2 before D3 ✓ (Afternoon < Night)
- One doctor per shift ✓
- All doctors assigned ✓

All constraints are satisfied.

Final Schedule

Doctor Shift

D1 Morning

D2 Afternoon

D3 Night

Search Tree

Start

|

D1 = Morning

|

D2 = Afternoon

|

D3 = Night

|

Goal Reached

Result

A valid schedule is obtained:

- **D1 → Morning**
 - **D2 → Afternoon**
 - **D3 → Night**
-

Question 2 – Solution

Grid Representation

```
S . . X .  
. X . X .  
. . . . .  
X X . X .  
. . . . G
```

Where:

- **S** = Start
 - **G** = Goal
 - **X** = Obstacle
 - **.** = Free cell
-

Manhattan Distance

$$h(n) = |x_1 - x_2| + |y_1 - y_2|$$

BFS Node Expansion

Step Expanded Node Queue After Expansion

1	S	(0,1), (1,0)
2	(0,1)	(1,0), (0,2)

Step Expanded Node Queue After Expansion

3	(1,0)	(0,2), (2,0)
4	(0,2)	(2,0), (1,2)
5	(1,2)	(2,0), (2,2)
6	(2,2)	(2,3), (3,2)
7	(2,3)	(2,4)
8	(2,4)	(3,4)
9	(3,4)	G
10	G	Goal Found

Optimal Path

S

↓

(0,1)

↓

(0,2)

↓

(1,2)

↓

(2,2)

↓

(2,3)

↓

(2,4)

↓

(3,4)

↓

G

Path Cost

Each move = 1

Number of moves = 8

Total Cost = 8

Result

- **Optimal Path:** $S \rightarrow (0,1) \rightarrow (0,2) \rightarrow (1,2) \rightarrow (2,2) \rightarrow (2,3) \rightarrow (2,4) \rightarrow (3,4) \rightarrow G$
 - **Total Cost = 8**
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Question 3 – Solution

(a) Constraint Analysis

Constraint	Effect on Robot
Move only Up, Down, Left, Right	Robot cannot move diagonally.
Blocked cells	Robot must avoid obstacles and choose another path.
Limited sensing	Robot observes only adjacent cells, making the environment partially observable.
Movement cost = 1	Robot tries to minimize the number of moves.
Risky zone cost = 3 (1 + 2)	Robot prefers safer paths with lower total cost.
Dynamic updates	Robot updates its map and replans whenever new obstacles are discovered.

(b) Solution Approach

Search Strategy

Uniform Cost Search (UCS)

Steps

1. Start from node S.
2. Insert S into the priority queue with cost 0.

3. Expand the node with the lowest cumulative cost.
 4. Ignore blocked cells.
 5. Add movement cost and any additional risky-zone cost.
 6. Update the priority queue with new cumulative costs.
 7. Continue until the goal node **G** is reached.
 8. Return the least-cost path.
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(c) AI Concepts

Intelligent Agent

The rescue robot is an intelligent agent because it senses its environment, processes information, and selects actions to safely reach the survivor.

Rationality

The robot is rational because it always chooses the action that minimizes the total travel cost while avoiding dangerous areas.

Environment Type

- Partially Observable
 - Dynamic
 - Sequential
 - Discrete
 - Single-Agent
-

(d) Justification

Uniform Cost Search is the most suitable algorithm because:

- It guarantees the least-cost path.
 - It considers different movement costs.
 - It avoids costly risky zones whenever possible.
 - It can adapt to newly discovered information during navigation.
 - It ensures safe and efficient rescue operations in a partially observable environment.
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Result

The rescue robot successfully reaches the survivor by using **Uniform Cost Search**, minimizing travel cost, avoiding obstacles and risky areas, and dynamically updating its knowledge as the environment changes.